

EE 210L: Digital Circuits and Logic Design Lab

Lab 5: Adders and Comparators

Name: _____

Date: _____

Partner(s): _____

Section: _____

Part 5.1: Half Adder

A	B	V at S (V)	S	V at C (V)	C	Expected S	Expected C
0	0						
0	1						
1	0						
1	1						

Part 5.2: Full Adder

Draw the full adder circuit you built. Label every gate and pin.

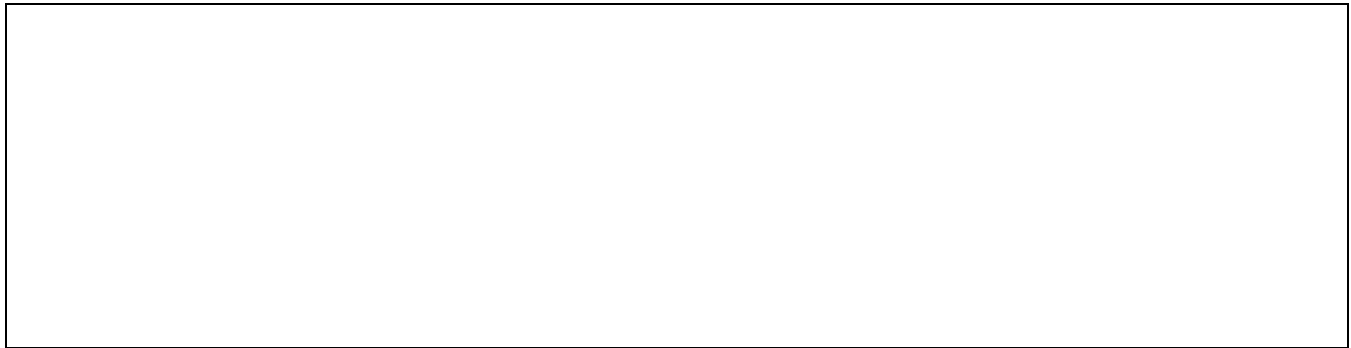


Figure 1: Full adder built from gates.

A	B	C _{in}	V at S (V)	S	V at C _{out} (V)	C _{out}	Expected S	Expected C _{out}
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

Part 5.3: Four-Bit Addition (74LS283)

A (dec)	A (bin)	B (dec)	B (bin)	$\Sigma 4 \Sigma 3 \Sigma 2 \Sigma 1$	C4	Sum as decimal	4-bit result correct?
9	1001	6	0110				
9	1001	11	1011				

How does C4 tell you that a 4-bit sum has overflowed?

Part 5.4: Adder/Subtractor

Draw how the XOR gates and the control line connect to the 74LS283.

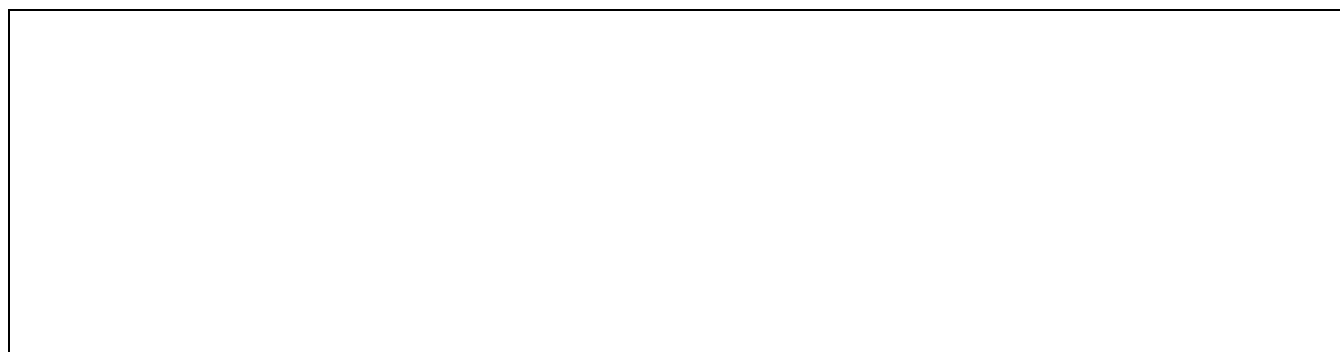


Figure 2: Adder/subtractor built around the 74LS283.

Control	Operation	A	B	$\Sigma 4 \Sigma 3 \Sigma 2 \Sigma 1$	C4	Result as decimal	Expected
0	A + B	7	3				
1	A – B	7	3				
1	A – B	12	5				
1	A – B						

What does C4 mean when the circuit is subtracting?

Part 5.5: Magnitude Comparator (74LS85)

Cascade inputs for standalone operation: pin 4 = _____, pin 2 = _____, pin 3 = _____

A (bin)	B (bin)	A>B out	A=B out	A<B out	Correct relation?

5.5.1 Cascade Inputs Set Incorrectly

With pin 3 tied to GND and $A = B$, the outputs read:

$A > B =$ _____ $A = B =$ _____ $A < B =$ _____

Explain why the comparator behaves this way, and what the cascade inputs are actually for.

Discussion and Conclusions

Address at least: why a half adder cannot be used for any bit but the least significant, what ripple carry costs you as the adder gets wider, and how one chip manages to both add and subtract.
